Bijectivity and the double angle formula

November 8, 2026

Injection, surjection, and bijection

Let us repeat the definitions from class. Suppose $f : X \mapsto Y$ is a map from X to Y.

Def: f is **injective** if
$\forall\ a, b \in X,\ a \neq b \implies f(a) \neq f(b)$
This is also called a **one-to-one** map.
Def: f is **surjective** if
$\forall y \in Y \exists x \in X$ such that $f(x) = y$ This is also called an **onto** map.
Def: f is **bijective** if it is bother injective and surjective

1. Draw a picture to illustrate a map which is injective but not surjective
2. List the asymptotes of the transformed function, and the intercepts
3. On a new graph, graph
$$\frac{1}{\sin(x/2)}$$
Explain how s graph is related to the cotangent graph above.